



SMART TOUCHLESS DUSTBIN

Ultrasonic sensing and servo-based automatic lid control using Arduino UNO

Project Type	Personal Project
Prepared By	Lalan Kumar Das
Roll No.	230108028
Controller	Arduino UNO
Sensor	HC-SR04 Ultrasonic Sensor
Actuator	Servo Motor

A compact embedded-system project that detects a nearby person without physical contact and automatically opens the dustbin lid. The controller uses a distance threshold and hysteresis to keep the servo stable when the measured distance fluctuates around the threshold.

TABLE OF CONTENTS

1. Project Overview
2. Objectives
3. Hardware and Software
4. System Architecture and Block Diagram
5. Circuit / Schematic Diagrams
6. Working Principle
7. Chattering Effect and Its Reduction
8. Arduino UNO Code
9. Testing, Result and Conclusion

PROJECT SNAPSHOT

Detection	HC-SR04 ultrasonic distance measurement
Control	Arduino UNO compares measured distance with thresholds
Action	Servo rotates to open the lid when a person is near
Reset	Servo returns to the initial position after the person moves away
Stability	Hysteresis/deadband reduces repeated open-close movement

1. PROJECT OVERVIEW

The Smart Touchless Dustbin is an Arduino-based automation project designed to open a dustbin lid without requiring physical contact. An HC-SR04 ultrasonic sensor detects the distance of a nearby person. Arduino UNO processes this measurement and commands a servo motor to rotate the lid mechanism. When the person moves away, the servo returns the lid to its initial position.

The project demonstrates a simple sensing and actuation workflow: sense → decide → actuate. A small hysteresis band is included so that sensor fluctuations do not make the servo repeatedly switch between positions.

2. OBJECTIVES

- Detect a person or hand without physical contact.
- Use an ultrasonic sensor for distance-based detection.
- Control the dustbin lid using a servo motor.
- Implement the system with Arduino UNO and simple embedded C++ code.
- Reduce servo chattering caused by small fluctuations in ultrasonic readings.

3. HARDWARE AND SOFTWARE

Component	Purpose
Arduino UNO	Main controller; reads the sensor and controls the servo
HC-SR04	Measures the distance of the approaching object/person
Servo motor	Provides controlled rotation for opening and closing the lid
5 V supply / USB	Provides power for the controller and circuit
Jumper wires	Electrical interconnections
Dustbin + lid mechanism	Mechanical structure for the final application
Arduino IDE	Code development and upload

4. SYSTEM ARCHITECTURE AND BLOCK DIAGRAM

The system is organized into five stages. The ultrasonic sensor measures distance, Arduino UNO makes the control decision, the servo provides mechanical movement, and the lid responds to the servo position.

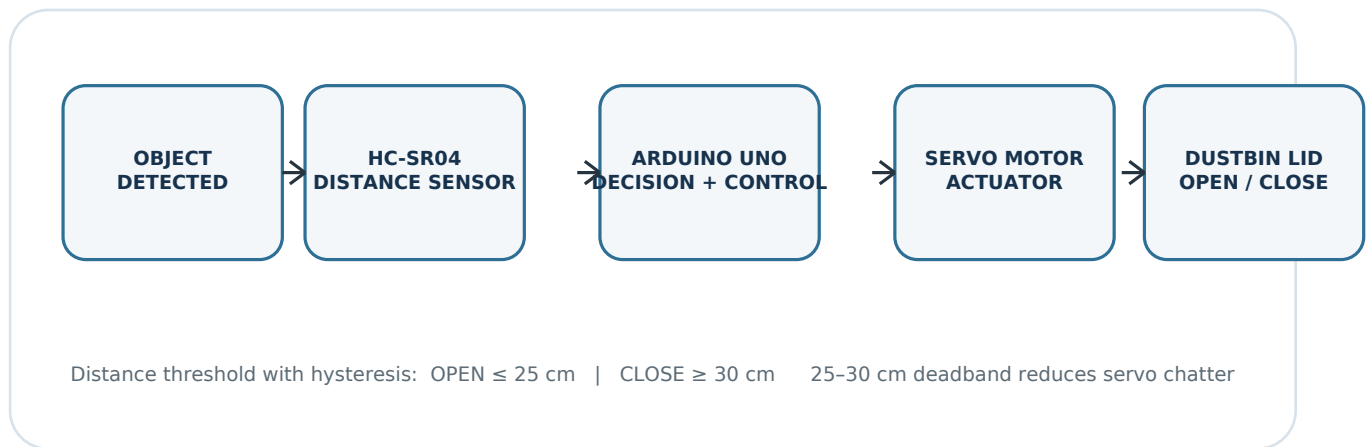


Figure 1. Functional block diagram of the touchless dustbin control system.

Control Logic

- Distance $\leq$ 25 cm: object/person is considered present → servo moves to the open position.
- Distance $\geq$ 30 cm: object/person is considered away → servo returns to the initial position.
- 25–30 cm: previous state is maintained → this deadband prevents rapid switching.

5. CIRCUIT / SCHEMATIC DIAGRAMS

The two supplied diagrams below document the electronic design: the schematic view and the practical wiring illustration.

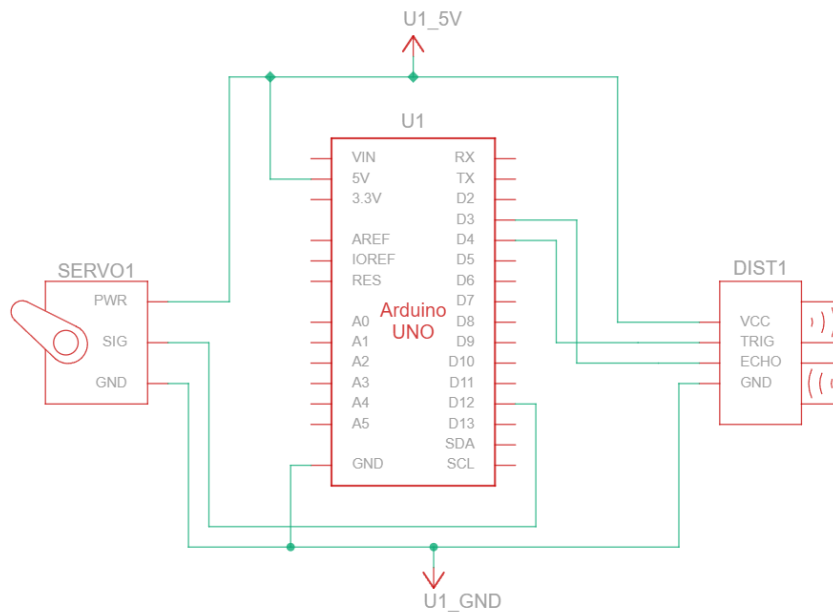


Figure 2. Schematic representation of the Arduino, ultrasonic sensor and servo motor.

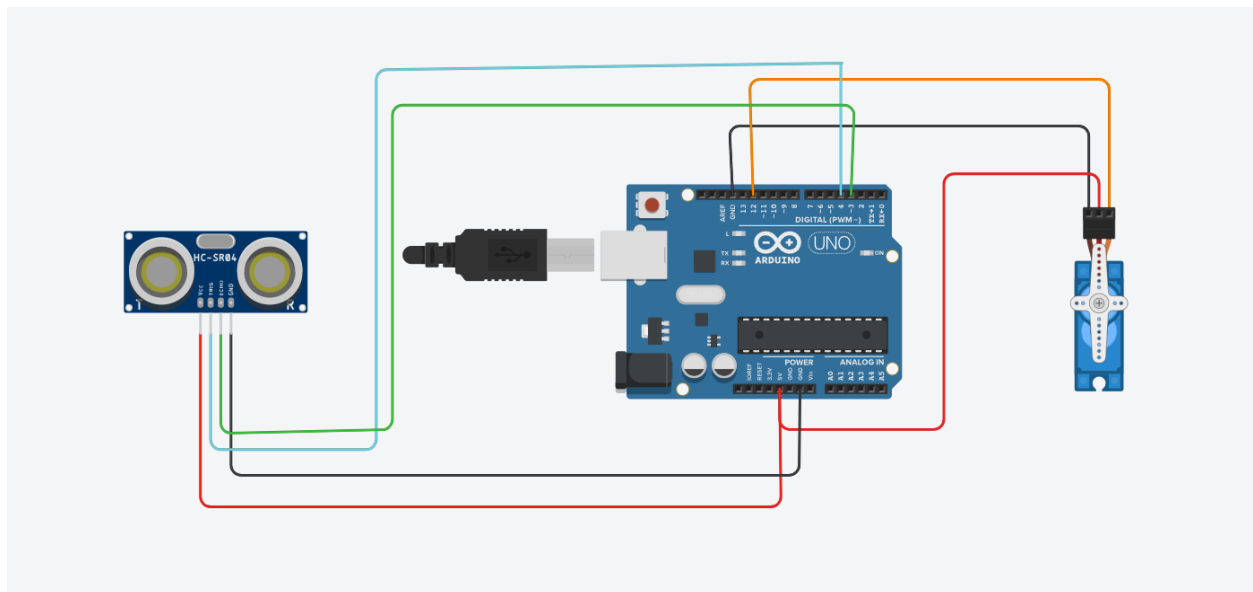


Figure 3. Wiring/connection illustration used for the prototype.

6. WORKING PRINCIPLE

	The HC-SR04 sends an ultrasonic pulse toward the nearby object.
	The echo time is measured by Arduino using pulseIn() .
	The measured time is converted into distance using the approximate speed of sound.
	Arduino compares the distance with the programmed detection thresholds.
	If the object is close enough, the servo rotates to open the lid.
	After the object moves sufficiently away, the servo returns to the initial position.

7. CHATTERING EFFECT AND ITS REDUCTION

Chattering occurs when the measured distance repeatedly moves around a single threshold because of sensor noise, movement, or small measurement variations. If the limit were exactly 25 cm, readings such as 24.8, 25.2, 24.9 and 25.1 cm could make the servo switch repeatedly.

The program therefore uses hysteresis: the lid opens at 25 cm or less, but closes only when the measured distance reaches 30 cm or more. The 25–30 cm region is a deadband that prevents rapid switching.

Result: smoother operation, fewer unnecessary servo movements, and better mechanical stability.

8. ARDUINO UNO CODE

Cleaned-up Arduino UNO program using hysteresis for stable touchless lid control.

```
#include <Servo.h>

Servo lidServo;

const int trigPin = 4;
const int echoPin = 3;
const int servoPin = 12;

const float OPEN_DISTANCE = 25.0; // cm
const float CLOSE_DISTANCE = 30.0; // cm

long duration;
float distanceCm;
bool lidOpen = false;

void setup()
{
  lidServo.attach(servoPin);
  lidServo.write(0);    // Initial/closed position

  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  Serial.begin(9600);
}

void loop()
{
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);

  duration = pulseIn(echoPin, HIGH, 30000);
  distanceCm = duration * 0.034 / 2.0;

  Serial.print("Distance: ");
  Serial.print(distanceCm);
  Serial.println(" cm");

  // Hysteresis control
  if (distanceCm <= OPEN_DISTANCE && !lidOpen)
  {
    lidServo.write(90);    // Open lid
    lidOpen = true;
  }
  else if (distanceCm >= CLOSE_DISTANCE && lidOpen)
  {
    lidServo.write(0);    // Close lid
    lidOpen = false;
  }

  delay(100);
}
```

9. TESTING, RESULT AND CONCLUSION

Testing Approach

Condition	Expected response
Object/person within 25 cm	Servo moves to 90° and lid opens
Object remains near the sensor	Lid stays open; servo is not repeatedly commanded
Distance between 25 and 30 cm	Previous lid state is maintained
Object moves to 30 cm or farther	Servo returns to 0° and lid closes

Result

The prototype demonstrates automatic, contactless lid operation using a low-cost ultrasonic sensor and servo motor. Hysteresis improves practical behavior by preventing unnecessary open-close switching near the detection boundary.

Conclusion

This personal project provides a practical introduction to embedded automation, sensor interfacing, actuator control and basic control-logic design. It also shows how a simple software technique can make a physical system more stable and usable.

Possible Future Improvements

- Add filtered/averaged distance measurement for smoother sensing.
- Use a dedicated servo power source if the mechanical lid requires higher current.
- Add an LED or buzzer for system-status indication.
- Use a stronger mechanical linkage and enclosed electronics for a finished product.

Project Author: Lalan Kumar Das | Roll No.: 230108028 | Project: Smart Touchless Dustbin